

Technical Appendix: MLB Stadium Weather Impact Dashboard

A. Data Sources

Source	Description	Access Method
Meteostat	Daily weather observations (temperature, wind, pressure)	meteostat Python library — Point + Daily API
pybaseball	Statcast pitch-level data; team game logs	statcast(), schedule_and_record(), team_game_logs()
annual_park_factor.xlsx	FanGraphs park factors by year (2021–2025)	Local Excel file, read via pd.read_excel()

Coverage: 2021–2025 regular seasons, all 30 MLB stadiums (home games).
Final dataset: Home games across all 30 parks after merging weather, game logs, and Statcast data.

B. Data Pipeline

B.1 Weather Ingestion

Weather is pulled at the game-day level for each stadium using its GPS coordinates. All 30 stadium locations are stored in a dictionary and queried in a loop:

```
for name, (lat, lon) in stadiums.items():
    point = Point(lat, lon)
    df = Daily(point, start, end).fetch()
```

Raw Meteostat fields used:

Field	Units	Description
tavg	°C	Average daily temperature
tmin / tmax	°C	Daily min/max temperature
wspd	km/h	Average wind speed
pres	hPa	Sea-level barometric pressure

B.2 Derived Variables

Temperature (Fahrenheit):

```
temp_f = (tavg × 9/5) + 32
```

Wind speed (mph):

```
wspd_mph = wspd × 0.621371
```

Air density (ideal gas law, kg/m³):

```
air_density = (pres × 100) / (287.05 × (tavg + 273.15))
```

where 287.05 J/(kg·K) is the specific gas constant for dry air. Air density varies meaningfully across stadiums — most notably at Coors Field (5,200 ft elevation) vs. sea-level parks.

Season of year (meteorological):

```
spring = Mar-May | summer = Jun-Aug | fall = Sep-Nov | winter = Dec-Feb
```

B.3 Game Log Merge

Game logs (runs scored R, runs allowed RA, strikeouts) are joined to weather observations on date and stadium. Games missing weather observations within ±1 day are dropped.

B.4 Park Factor Adjustment

Annual park factors from FanGraphs are merged by year and stadium. Residual runs are computed as:

```
residual_runs = R - (league_avg_runs × park_factor / 100)
residual_RA   = RA - (league_avg_runs × park_factor / 100)
```

C. Variable Definitions

Variable	Type	Definition
R	Game outcome	Home team runs scored
RA	Game outcome	Opponent runs allowed
total_runs	Game outcome	R + RA
SO	Game outcome	Total strikeouts (both teams)
park_factor	Control	FanGraphs annual park factor (100 = neutral)
residual_runs	Derived	Runs scored minus park-adjusted expectation
temp_f	Weather	Game-day average temperature (°F)
wspd_mph	Weather	Game-day average wind speed (mph)
pres	Weather	Barometric pressure (hPa)
air_density	Weather	Derived air density (kg/m³)
roof_status	Contextual	Open / Closed (from game log)
region	Contextual	Stadium region (Northeast / North / South / West)

Variable	Type	Definition
is_extreme	Flag	z-score > 2.5 on any weather variable

D. Extreme Weather Classification

Type	Threshold
Extreme cold	temp_f in bottom 5th percentile (per stadium)
Extreme heat	temp_f in top 95th percentile (per stadium)
High wind	wspd_mph in top 95th percentile (per stadium)
Low pressure	pres in bottom 5th percentile (per stadium)

E. Analysis Phases

Phase 1 — Park Structure Baseline

Establishes each stadium's physical and environmental characteristics as fixed controls, including outfield dimensions, wall height, elevation, and annual park factor. Key contrasts include:

Stadium	Elevation	Avg Park Factor	Climate Type
Coors Field (COL)	5,200 ft	~115 (hitter-friendly)	Semi-arid, high altitude
T-Mobile Park (SEA)	0 ft	~92 (pitcher-friendly)	Marine, mild
Chase Field (ARI)	1,100 ft	~100 (neutral)	Desert, retractable roof
Fenway Park (BOS)	20 ft	~104 (hitter-friendly)	Humid continental

Phase 2 — Weather Characterization

Describes each stadium's observed climate profile across the 2021–2025 seasons, including seasonal temperature curves, wind distributions, and extreme event frequency. Stadiums are grouped into four regions:

- Northeast: Fenway Park, Yankee Stadium, Citi Field, Citizens Bank Park, Nationals Park, PNC Park
- North: Wrigley Field, Guaranteed Rate Field, Comerica Park, Progressive Field, Target Field, American Family Field, Rogers Centre, Camden Yards
- South: Minute Maid Park, Truist Park, Globe Life Field, Kauffman Stadium, Busch Stadium, Great American Ball Park, LoanDepot Park, Tropicana Field
- West: Dodger Stadium, Oracle Park, Coors Field, Petco Park, Chase Field, T-Mobile Park, Angel Stadium, RingCentral Coliseum

Phase 3 — Performance Analysis

Runs and strikeouts are analyzed across weather tertiles (cool / moderate / warm for temperature; low / moderate / high for wind) league-wide and by region. Linear correlations and residual analyses (post park-factor adjustment) are reported for each stadium and in aggregate.

Phase 4 — Year-over-Year Park Factor Analysis

Annual park factors are correlated with seasonal weather profiles across all stadiums to evaluate whether year-to-year variability in park factors is partially explained by weather patterns. Regional trends and outlier years are identified.

F. Visualization Index

Figure ID	File	Description
1.1	viz_1_1_monthly_temperature.png	Monthly average temperature across stadiums (2021–2025)
1.2	viz_1_2_temperature_distribution.png	Temperature distribution histogram by region
1.3	viz_1_3_wind_distribution.png	Wind speed distribution by stadium
1.4	viz_1_4_extreme_weather_by_year.png	Extreme weather game counts by year
1.5	viz_1_5_extreme_weather_types.png	Extreme event breakdown by type and region
1.6	viz_1_6_day_night_by_month.png	Day vs. night game distribution
1.7	viz_1_7_roof_status_by_month.png	Roof open/closed frequency by month
2.1	viz_2_1_temperature_runs.png	Temperature vs. total runs (scatter + trend)
2.2	viz_2_2_temperature_strikeouts.png	Temperature vs. strikeouts
2.3	viz_2_3_day_night_runs.png	Runs by day/night game type
2.4	viz_2_4_day_night_strikeouts.png	Strikeouts by day/night game type
2.6	viz_2_6_residual_runs_boxplot.png	Park-adjusted residual runs by temp tertile
3.1	viz_3_1_park_factor_trend.png	Park factor trends across stadiums 2021–2025
3.2	viz_3_2_annual_weather_performance.png	Annual weather profile vs. performance by stadium
3.3	viz_3_3_correlation_heatmap.png	Weather–performance correlation matrix
3.4	viz_3_4_2024_outlier_comparison.png	2024 outlier stadiums vs. baseline
3.5	viz_3_5_parkfactor_vs_extreme_scatter.png	Park factor vs. extreme weather frequency

G. Software & Libraries

Library	Purpose
pandas	Data wrangling and merging
numpy	Numerical computation, z-scores
matplotlib	Base plotting
seaborn	Statistical visualizations
meteostat	Historical weather data retrieval
pybaseball	MLB Statcast and game log data

H. Limitations

1. Weather is game-day daily average, not first-pitch conditions. Retractable-roof games during rain may show misleadingly mild ambient weather.
2. Statcast merge completeness: Statcast coverage is incomplete for some 2021 games due to COVID-shortened scheduling artifacts.
3. Extreme weather z-scores are stadium-relative: A "normal" day at Coors Field may be extreme by another park's standards; cross-stadium comparisons should account for baseline differences.
4. Confounders not modeled: Pitcher quality, opponent strength, and bullpen usage are not controlled for in weather-performance regressions.
5. Meteostat data gaps: A small number of stadiums have missing pressure or wind readings for certain dates; affected games are excluded from those specific analyses.